APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS

UNIT – V

Standard Variational Methods

Syllabus (as given)

- Standard variational methods: linear fluid flow problems, steady state problems, transient problems.

1. Introduction — Variational Principle and Weak Form

Variational methods obtain the approximate solution of a differential equation by (i) minimising an integral quantity called a functional, or (ii) forcing the weighted integral of the residual to zero. They are the mathematical foundation of the finite element method (FEM).

Functional and Euler–Lagrange equation

A functional $I[u]$ assigns a number to a function $u(x)$. Example (1-D Poisson / heat conduction / Poiseuille flow):

$$I[u] = \int_0^L [\tfrac{1}{2} k (du/dx)^2 - f u] \, dx$$

Among all admissible functions (satisfying essential BCs), the true solution makes I stationary: $\delta I = 0$

-> Euler–Lagrange equation:

$$-d/dx(k \, du/dx) = f \qquad \text{(differential equation)}$$

- Essential (Dirichlet) BCs: $u = \bar{u}$ on boundary — must be satisfied by the trial function.
- Natural (Neumann) BCs: $k \, du/dx = q$ — arise automatically from the variation; need not be imposed on the trial function.
- For self-adjoint (symmetric) operators the functional exists and gives symmetric stiffness matrices; for fluid flow with convection (non-self-adjoint) we use Galerkin weighted residuals instead.

Analogy

- Structural mechanics: functional = total potential energy; minimum potential energy principle.
- Heat conduction / Stokes flow: functional = dissipation-type integral; Stokes flow minimises viscous dissipation (Helmholtz minimum dissipation theorem).

2. Rayleigh–Ritz Method

Assume a trial solution as a finite series of known basis functions:

$$u(x) \approx u_h(x) = \sum_{j=1}^{n} c_j \varphi_j(x) \quad (\varphi_j \text{ satisfy the essential BCs})$$

Substitute in the functional $I(c_1, \ldots, c_n)$ and require stationarity: $\partial I/\partial c_i = 0$, $i = 1 \ldots n$ -> matrix equation

$$\sum_j K_{ij} c_j = F_i, \quad K_{ij} = \int k \varphi_i' \varphi_j' \, dx, \quad F_i = \int f \varphi_i \, dx$$

Example — Ritz solution

$-u'' = 1$ on $0 < x < 1$, $u(0) = u(1) = 0$. Trial $u = c\,x(1 - x)$ (satisfies BCs).

$I = \int [\tfrac{1}{2}(c(1 - 2x))^2 - c\,x(1 - x)]\, dx = c^2/6 - c/6$ -> $dI/dc = c/3 - 1/6 = 0$ -> $c = 1/2$.

$u = x(1 - x)/2$ which is the exact solution (the trial space contained it).

3. Method of Weighted Residuals (MWR)

When no functional exists (e.g. convection, Navier–Stokes), use the residual of the differential equation $L(u) = f$:

$$R(x) = L(u_h) - f \neq 0 \quad \text{->} \quad \int_\Omega W_i(x) R(x)\, dx = 0, \quad i = 1 \ldots n$$

Method	Weight function W_i
Collocation	Dirac delta $\delta(x - x_i)$ ($R = 0$ at chosen points)
Subdomain	1 on subdomain Ω_i, 0 elsewhere (average $R = 0$)
Least squares	$W_i = \partial R/\partial c_i$ (minimise $\int R^2\, dx$)
Moments	$W_i = x^{i-1}$
Galerkin	$W_i = \varphi_i$ (same as basis/shape function)

- Galerkin method gives symmetric matrices for self-adjoint operators and, in that case, equals the Ritz method; it is the standard choice in FEM.

Weak (variational) form of the 1-D problem

Multiply $-(k u')' = f$ by test function w ($w = 0$ at essential BC) and integrate by parts:

$$\int_0^L k w' u' \, dx = \int_0^L f w \, dx + [\, w k u' \,]_0^L$$

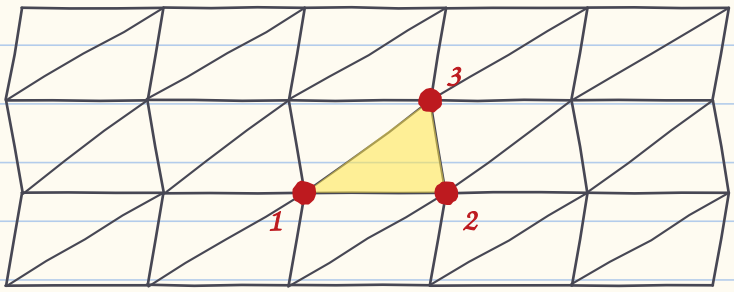
- Integration by parts lowers the derivative order (C^0 shape functions suffice) and brings in natural BCs through the boundary term.

Galerkin example

Same problem $-u'' = 1$, trial $u = a\,x(1-x)$, $w = x(1-x)$: $\int (u'w' - w)dx = 0$.

$\int (a(1-2x)^2 - x(1-x))dx = a/3 - 1/6 = 0 \Rightarrow a = 1/2$. √

4. Finite Element Method — Discretisation

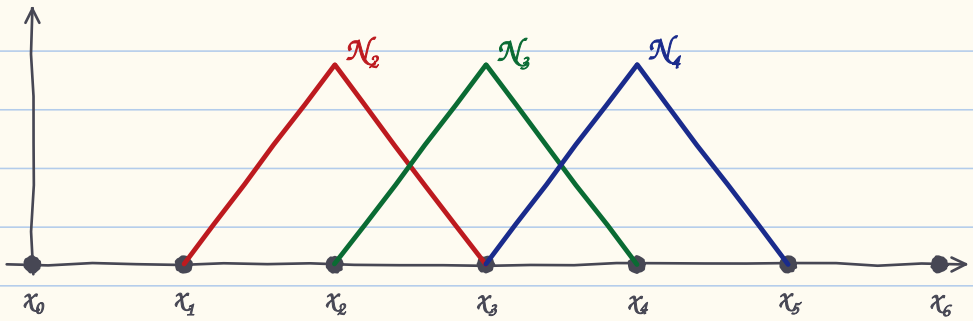


Element e (nodes 1,2,3)

Nodes = unknowns u_j

Mesh = assembly of elements

1. Divide the domain into elements connected at nodes (1-D: lines; 2-D: triangles/quads; 3-D: tets/hexes).
2. Choose local shape (interpolation) functions $N_j(x)$: $u^e(x) = \Sigma\, N_j^e(x)\, u_j$.
3. Apply Galerkin weighted residual / variational statement to every element → element matrices.
4. Assemble element matrices into the global system; impose boundary conditions; solve $K u = F$.
5. Post-process: gradients, fluxes, velocities, error estimates.



$N_i(x_i) = 1$

$N_i(x_j) = 0$

$(i \neq j)$

- Global "hat" functions: $N_i = 1$ at node i, 0 at all other nodes — compact support, so the matrix is sparse and banded.

1-D linear element (length $h = x_2 - x_1$)

$$N_1 = (x_2 - x)/h\,, \quad N_2 = (x - x_1)/h \qquad (N_1 + N_2 = 1)$$

$$K^e = (k/h)\,[\,1\ \ -1\,;\,-1\ \ 1\,]\,, \qquad F^e = (f\,h/2)\,[\,1\,;\,1\,]\,, \qquad M^e = (\rho\,h/6)\,[\,2\ \ 1\,;\,1\ \ 2\,]$$

$K^e_{ij} = \int k N_i' N_j'\, dx$, $F^e_i = \int f N_i\, dx$, $M^e_{ij} = \int \rho N_i N_j\, dx$ (consistent mass matrix).

2-D linear triangular element

$$N_i = (a_i + b_i x + c_i y)/(2A), \quad b_1 = y_2 - y_3,\ c_1 = x_3 - x_2 \text{ (cyclic)}$$

$$K^e_{ij} = (k/4A)(b_i b_j + c_i c_j), \quad F^e_i = fA/3, \quad M^e_{ij} = (\rho A/12)(1 + \delta_{ij})$$

A = area of triangle. Quadrilateral elements: isoparametric mapping with natural coordinates (ξ, η) and Gauss quadrature $\int f\, d\xi \approx \Sigma\, w_g f(\xi_g)$.

Quadratic element and natural coordinates (1-D)

Use $\xi \in [-1, 1]$ with $x = \Sigma\, N_i(\xi) x_i$ (isoparametric). Jacobian $J = dx/d\xi = h/2$ for a straight element.

$$\text{Linear: } N_1 = (1 - \xi)/2, \quad N_2 = (1 + \xi)/2$$

$$\text{Quadratic (3 nodes at } \xi = -1, 0, 1\text{): } N_1 = \xi(\xi - 1)/2, \quad N_2 = 1 - \xi^2, \quad N_3 = \xi(\xi + 1)/2$$

$$K^e_{ij} = \int_{-1}^{1} k\,(dN_i/d\xi)(dN_j/d\xi)(1/J)\, d\xi, \quad F^e_i = \int_{-1}^{1} f N_i J\, d\xi$$

Gauss points n	Locations ξ_g	Weights w_g
1	0	2
2	±0.57735	1, 1
3	0, ±0.77460	8/9, 5/9, 5/9

- n-point Gauss quadrature is exact for polynomials of degree $2n - 1$. Quadratic stiffness matrix: $K^e = (k/3h)[7\ -8\ 1;\ -8\ 16\ -8;\ 1\ -8\ 7]$.
- Quadratic elements give $O(h^3)$ in L^2 and $O(h^2)$ in energy norm — much higher accuracy for the same number of nodes.

5. Linear Fluid Flow Problems

"Linear" = governing equations are linear in the unknowns: potential flow, creeping (Stokes) flow, fully developed laminar duct flow, unsteady diffusion of momentum/vorticity — no convective $(u \cdot \nabla u)$ term.

5.1 Potential (irrotational, incompressible) flow

$$\nabla^2 \varphi = 0 \text{ in } \Omega, \quad u = \partial\varphi/\partial x,\ v = \partial\varphi/\partial y; \quad \varphi = \varphi_0\ (\Gamma_1), \quad \partial\varphi/\partial n = u_n\ (\Gamma_2)$$

$$I[\varphi] = \tfrac{1}{2} \int_\Omega (\varphi_x^2 + \varphi_y^2)\, d\Omega - \int_{\Gamma_2} u_n \varphi\, d\Gamma$$

- Stream function version $\nabla^2 \psi = 0$ (ψ = const on solid walls). Element equations are those of Laplace/Poisson: $K_{ij} = \int \nabla N_i \cdot \nabla N_j d\Omega$.

5.2 Fully developed laminar flow in a duct (Poiseuille / Poisson problem)

$$\mu \nabla^2 w = dp/dz = -G, \quad w = 0 \text{ on walls}$$

$$I[w] = \int_\Omega [\, \tfrac{1}{2}\mu (w_x^2 + w_y^2) - G\, w \,]\, d\Omega$$

Same Poisson form as heat conduction with heat source: $K_{ij} = \mu \int \nabla N_i \cdot \nabla N_j d\Omega$, $F_i = G \int N_i d\Omega$.

Volume flow rate $Q = \int w\, d\Omega$; friction factor from the wall gradient.

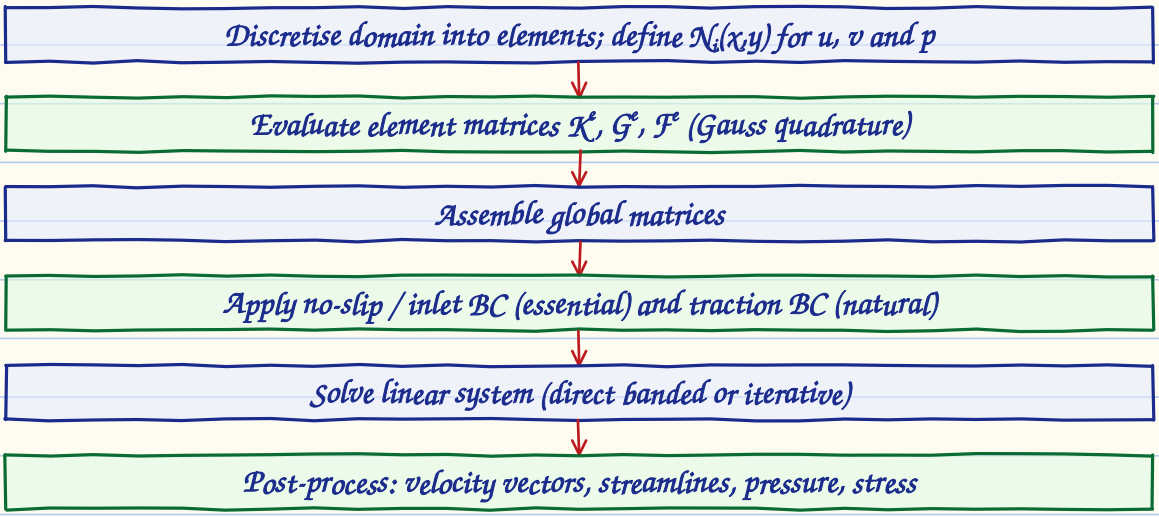
5.3 Stokes (creeping) flow — mixed formulation

$$-\mu \nabla^2 \mathcal{V} + \nabla p = f, \quad \nabla \cdot \mathcal{V} = 0$$

Weak form: $\int \mu\, \nabla w : \nabla \mathcal{V}\, d\Omega - \int p\, \nabla \cdot w\, d\Omega = \int w \cdot f\, d\Omega$; $\int q\, \nabla \cdot \mathcal{V}\, d\Omega = 0$

Matrix system: $[\, K \;\; -G^T \;;\; G \;\; 0 \,]\, [\, \mathcal{V} \;;\; P \,] = [\, F \;;\; 0 \,]$

- **LBB (Babuška–Brezzi) condition:** velocity and pressure spaces cannot be chosen arbitrarily (equal-order linear elements give spurious pressure modes). Stable pairs: Taylor–Hood P2–P1 (quadratic velocity, linear pressure), Q2–Q1, MINI element.
- **Penalty method:** relax incompressibility $p = -\lambda\, \nabla \cdot \mathcal{V}$ ($\lambda \sim 10^6\, \mu$) and eliminate p: $(K + \lambda\, G^T M_p^{-1} G)\mathcal{V} = F$ — reduced integration of the penalty term avoids locking.
- **Stream function–vorticity:** biharmonic $\nabla^4 \psi = 0$ needs C^1 elements; or split into two Poisson equations.



Applying boundary conditions to $K\,u = F$

Method	How
Elimination	strike out rows/columns of known u_k; move $K_{ik} u_k$ to the right side
Penalty	$K_{kk} \leftarrow K_{kk} + C$, $F_k \leftarrow F_k + C\, \bar{u}_k$ with $C \sim 10^6 \max K_{ii}$
Lagrange multiplier	add constraint equation $u_k = \bar{u}_k$; enlarges the system
Natural BC	add boundary load q (or $h(T_\infty - T)$) to F_i and h to K_{ii} (Robin)

Worked example: 2-D triangle element

Element matrix for Laplace / Poisson (k = 1)

Triangle with nodes 1 (0,0), 2 (1,0), 3 (0,1): $A = 1/2$. $b_1 = y_2 - y_3 = -1$, $b_2 = y_3 - y_1 = 1$, $b_3 = y_1 - y_2 = 0$;

$c_1 = x_3 - x_2 = -1$, $c_2 = x_1 - x_3 = 0$, $c_3 = x_2 - x_1 = 1$.

$K_{11} = (b_1^2 + c_1^2)/4A = 2/2 = 1$, $K_{12} = (b_1b_2 + c_1c_2)/4A = -1/2$, $K_{13} = -1/2$, $K_{22} = 1/2$, $K_{23} = 0$, $K_{33} = 1/2$.

Row sums are zero (constant field gives zero gradient)√; matrix symmetric √.

Assembly of three 1-D elements (non-uniform k)

Assembly rule

Elements e1 (nodes 1-2, k_1, h_1), e2 (2-3), e3 (3-4). Global K (4×4) is tridiagonal:

$K_{11} = k_1/h_1$, $K_{22} = k_1/h_1 + k_2/h_2$, $K_{33} = k_2/h_2 + k_3/h_3$, $K_{44} = k_3/h_3$, $K_{12} = -k_1/h_1$, …

Each element contributes to the diagonal of its two nodes (adds) and to the off-diagonal between them — interfaces between materials / fluids with different μ are handled naturally.

6. Steady State Problems

Steady problems give a system $Ku = F$ with no time derivative. Worked 1-D example (axial heat conduction or Poiseuille flow between plates):

Example: $-u'' = 1$, $u(0) = u(1) = 0$, two linear elements ($h = 0.5$)

Nodes $x = 0, 0.5, 1$. Element matrices $K^e = (1/0.5)[1\ -1; -1\ 1] = [2\ -2; -2\ 2]$, $F^e = (0.5/2)(1, 1) = (0.25, 0.25)$.

Assembly: $K = [2\ -2\ 0; -2\ 4\ -2; 0\ -2\ 2]$, $F = (0.25, 0.5, 0.25)$.

Apply $u_1 = u_3 = 0$ (delete rows/columns 1, 3): $4u_2 = 0.5$ -> $u_2 = 0.125$.

Exact $u(0.5) = x(1 - x)/2 = 0.125$ √ (nodally exact for this 1-D problem).

Plane Poiseuille flow by FEM

Flow between plates at $y = \pm b$ with pressure gradient $G = -dp/dx$: $\mu w'' = -G$. With symmetry, solve on $0 \le y \le b$ with $w(b) = 0$ (essential) and $w'(0) = 0$ (natural, symmetric). Exact: $w = (G/2\mu)(b^2 - y^2)$, $w_{max} = Gb^2/2\mu$, mean velocity $= 2w_{max}/3$.

Steps for steady-state fluid problems

1. Weak form from governing equation (Galerkin).
2. Element matrices (shape functions, numerical integration).
3. Assembly: $K_{IJ} \mathrel{+}= K^{e}_{ij}$ where I, J = global numbers of local nodes i, j.
4. Essential BC: delete rows/columns, or penalty (large number on diagonal), or modify RHS.
5. Solve; recover derived variables (shear stress $\tau = \mu\, du/dy$, flux) — they are less accurate than u (one order lower).

- Convection-dominated steady flows (non-self-adjoint) show wiggles with plain Galerkin for large Peclet number -> use SUPG (streamline upwind Petrov–Galerkin): $W_i = N_i + (\alpha h/2\,|u|)u\cdot\nabla N_i$.
- Non-linear steady problems (Navier–Stokes): Picard (successive substitution) or Newton–Raphson iterations on $K(u)u = F$.

Convection–diffusion by Galerkin / SUPG (steady, 1-D)

$$u\, d\varphi/dx = \Gamma\, d^2\varphi/dx^2 \;\rightarrow\; \int (W u \varphi' + \Gamma W \varphi')\, dx = 0$$

Linear elements, uniform mesh, Galerkin gives at an interior node:

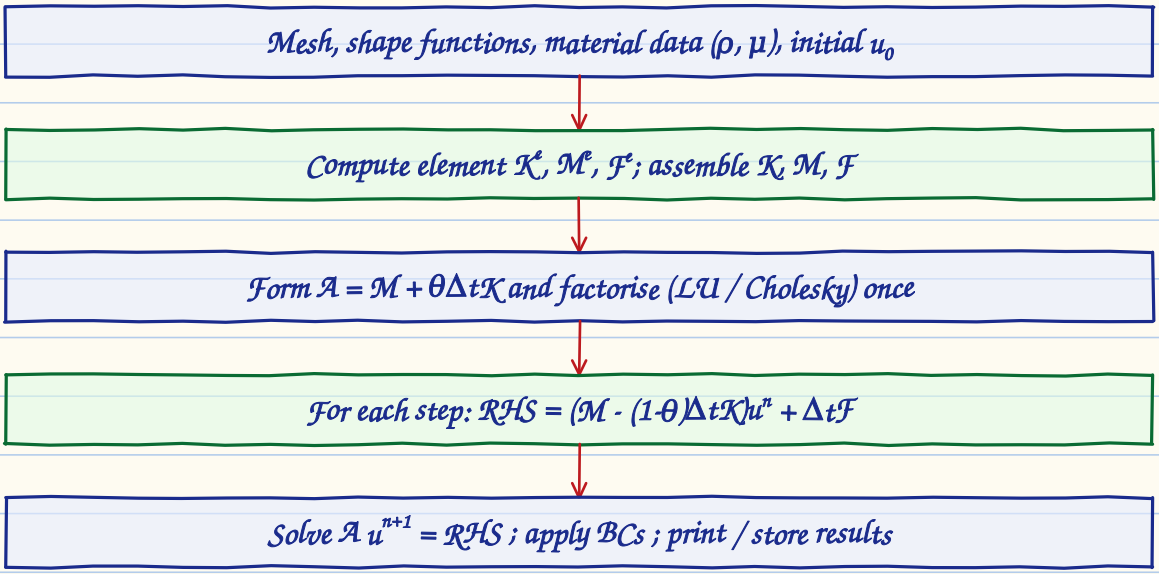
$(u/2)(\varphi_{i+1} - \varphi_{i-1}) = (\Gamma/h)(\varphi_{i+1} - 2\varphi_i + \varphi_{i-1})$ — identical to central differencing -> wiggles for $Pe_h = uh/2\Gamma > 1$.

- SUPG adds artificial diffusion only along the streamline: $\Gamma_{art} = \xi(Pe)\, u\, h/2$, $\xi = \coth(Pe) - 1/Pe$ (optimal 1-D value, gives nodally exact solution).
- Other stabilisations: GLS (Galerkin least squares), bubble functions, Taylor–Galerkin (time-dependent), characteristic Galerkin.

Pipe/duct flow and Couette flow examples

- Couette flow between plates (gap H, top plate speed U): $\mu w'' = 0$ with $w(0) = 0$, $w(H) = U$ -> linear profile; one linear element reproduces it exactly.
- Hagen–Poiseuille flow in a circular pipe: axisymmetric weak form $\int \mu\, (dN_i/dr)(dw/dr)\, r\, dr = \int G\, N_i\, r\, dr$; exact $w = (G/4\mu)(R^2 - r^2)$, $Q = \pi G R^4/8\mu$.
- Flow in a rectangular duct: 2-D Poisson on the cross-section, 4 triangular or quadrilateral elements per quadrant already gives Q within a few percent.
- Potential flow around a cylinder in a channel: solve $\nabla^2\psi = 0$ with $\psi = 0$ on the body and the free-stream $\psi = U y$ far away; pressure from Bernoulli $p = p_\infty + \tfrac{1}{2}\rho(U^2 - |\nabla\psi|^2)$.

Algorithm for a transient FEM flow solver



7. Transient Problems

Time-dependent linear problem, e.g. unsteady diffusion of momentum (Stokes first problem) $\rho\,\partial u/\partial t = \mu\,\partial^2 u/\partial y^2$ or heat conduction. Galerkin in space (semi-discretisation):

$$M\dot{u} + Ku = F(t) \qquad \text{(system of ODEs)}$$

where $M_{ij} = \int \rho N_i N_j\,dx$ (mass matrix), $K_{ij} = \int \mu\, N_i' N_j'\,dx$. Initial condition $u(0) = u_0$.

Time integration — θ-method (generalised trapezoidal)

$$(M + \theta\Delta t\,K)\,u^{n+1} = (M - (1-\theta)\Delta t\,K)\,u^n + \Delta t[\,\theta F^{n+1} + (1-\theta)F^n\,]$$

θ	Name	Stability	Accuracy
0	Forward Euler	$\Delta t \le 2/\lambda_{max}$	$O(\Delta t)$
½	Crank–Nicolson	unconditional	$O(\Delta t^2)$
2/3	Galerkin	unconditional	$O(\Delta t)$
1	Backward Euler	unconditional	$O(\Delta t)$

- λ_{max} = largest eigenvalue of $M^{-1}K \sim 4\mu/(\rho h^2)$ for linear elements: explicit limit $\Delta t \le \rho h^2/(2\mu)$ (for lumped mass $\Delta t \le \rho h^2/2\mu$).
- For $\theta \ge ½$ unconditionally stable; for $\theta < ½$ conditionally stable: $\Delta t \le 2/((1-2\theta)\lambda_{max})$.
- Lumped mass matrix $M_L = (\rho h/2)I$ (row sum) makes explicit schemes cheap (diagonal M); consistent M is more accurate for wave problems but can give oscillations with $\theta = 0$.
- Each step solves a linear system with the same matrix $(M + \theta\Delta t K)$ — factorise once, reuse; this is the great advantage for linear problems.

Alternatives

- Finite difference in time: Backward Euler / BDF2; Newmark-β for second-order systems $M\ddot{u} + C\dot{u} + Ku = F$.
- Space–time elements, discontinuous Galerkin in time; modal (eigenfunction) expansion $u = \Sigma a_k(t)\psi_k(x)$ giving uncoupled ODEs $\dot{a}_k + \lambda_k a_k = 0$.

Example: transient 1-D heat/diffusion with one free node

Rod $0 \le x \le 1$, two elements ($h = 0.5$), $u(0) = u(1) = 0$, $\rho = \mu = 1$, no source, $u(0.5, 0) = 1$.

Free node 2: $M_{22} = 2h/3 = 1/3$, $K_{22} = 4$ -> $(1/3)\dot{u} + 4u = 0$ -> $\dot{u} = -12u$, exact semi-discrete $u = e^{-12t}$ (true: $e^{-\pi^2 t} \approx e^{-9.87t}$).

Backward Euler: $u^{n+1} = u^n/(1 + 12\Delta t)$; Crank–Nicolson: $u^{n+1} = u^n(1 - 6\Delta t)/(1 + 6\Delta t)$;

Forward Euler: stable only if $12\Delta t \le 2$ -> $\Delta t \le 1/6$.

Transient fluid-flow applications

- Stokes first/second problem (impulsively moved plate, oscillating plate): diffusion of vorticity; boundary-layer thickness $\delta \sim \sqrt{(\nu t)}$.
- Start-up of Poiseuille flow in a channel/pipe: $u(y,t) \to$ steady parabola as $t \to \infty$; time scale $t \sim b^2/\nu$.
- Unsteady Stokes with incompressibility: saddle-point system solved at each time step by projection / mixed elements.
- Unsteady Navier–Stokes (non-linear): operator splitting, characteristic-Galerkin or SUPG for convection, Newton iterations per step.

Short answer questions

Q1. Difference between Ritz and Galerkin? -> Ritz needs a functional (self-adjoint problem); Galerkin works directly with the differential equation.

Q2. What is a shape function? -> interpolation function with $N_i = 1$ at node i, 0 at other nodes; $\Sigma N_i = 1$.

Q3. What are essential and natural BCs? -> essential: imposed on u; natural: appear in the weak form (fluxes/tractions).

Q4. Why integrate by parts? -> reduces required continuity (C^0) and introduces natural BCs.

Q5. What is the consistent vs lumped mass matrix? -> consistent: $\int \rho N_i N_j$; lumped: diagonal (row sum), used for explicit schemes.

Q6. Why Galerkin is poor for high Peclet flows? -> central-like, oscillations; use SUPG.

Q7. What is locking in penalty methods? -> too large λ makes the matrix singular, solved by reduced/selective integration.

Comparison: Ritz vs Galerkin vs collocation (example $-u'' = 1$)

Method	Weight / condition	Result with $u = a\,x(1 - x)$
Ritz	$\partial I/\partial a = 0$	$a = 1/2$
Galerkin	$W = x(1 - x)$	$a = 1/2$
Collocation at $x = ½$	$R(½) = 0$	$2a - 1 = 0$ -> $a = 1/2$
Least squares	$\int R^2$ min	$a = 1/2$

All methods give the exact solution here because the trial function belongs to the solution space; in general they differ.

7A. Beyond the Linear Case — Weak Form of Navier–Stokes (brief)

$$\int w\cdot(\rho\partial V/\partial t + \rho(V\cdot\nabla)V)\,d\Omega + \int \mu\,\nabla w:\nabla V\,d\Omega - \int p\,\nabla\cdot w\,d\Omega = \int w\cdot f\,d\Omega + \int_\Gamma w\cdot t\,d\Gamma, \quad \int q\,\nabla\cdot V\,d\Omega = 0$$

- Extra non-linear convective integral gives a non-symmetric, solution-dependent matrix $K(V)$; solve with Picard or Newton iteration (linearise $V\cdot\nabla V \approx V^k\cdot\nabla V^{k+1}$).
- Penalty / segregated (projection) / coupled solvers; time step limited by CFL for explicit convection.
- Vorticity–stream function FEM: two Poisson problems $\int\nabla N\cdot\nabla\psi\,d\Omega = \int N\omega\,d\Omega$ and vorticity transport solved with SUPG; wall vorticity from natural BC of ψ.

Element	Nodes / DOF	Use in fluid problems
2-node line	linear	1-D channel, pipe network
3-node triangle (T3)	linear	Laplace, Poisson, ψ, φ
6-node triangle (T6)	quadratic	velocity in Taylor–Hood
4-node quad (Q4)	bilinear	pressure, temperature
8/9-node quad	biquadratic	velocity, high accuracy
4-node tetrahedron	linear	3-D potential flow

Properties of the FEM matrices

- K is symmetric positive definite for self-adjoint problems with essential BCs; sparse banded -> Cholesky or conjugate-gradient solvers.
- Stokes saddle-point matrix is symmetric indefinite; solved with Uzawa iteration, Schur

complement or block preconditioned GMRES/MINRES.

- Bandwidth minimisation by good node numbering (Cuthill–McKee) reduces memory and solve time.

8. Error Estimates and Convergence

- Energy-norm error for elements of degree p: $||u - u_h||_E \le C h^p |u|_{p+1}$ and in L^2 norm $O(h^{p+1})$ (linear elements: 1st order in energy norm, 2nd order in L^2).
- Ritz/Galerkin solution is the best approximation to the exact solution in the energy norm (Céa's lemma).
- h-refinement (smaller elements), p-refinement (higher order), r-refinement (node movement), adaptive meshing with a-posteriori error estimators.

Aspect	FEM (variational)	FVM / FDM
Basic idea	minimise functional / weak form	flux balance / Taylor series
Geometry	excellent, unstructured	good (FVM) / structured (FDM)
BCs	natural BC built-in	imposed through fluxes/ghost nodes
Conservation	global weak sense	local exact (FVM)

Quick revision — Unit V

- Functional $I[u] = \int (\tfrac{1}{2} k u'^2 - f u)dx$; stationarity gives the differential equation + natural BC.
- Ritz: minimise I over trial space; Galerkin: weight function = shape function (works for any PDE).
- Element: $K^e = (k/h)[1\ -1;\ -1\ 1]$, $F^e = (fh/2)(1, 1)$, $M^e = (\rho h/6)[2\ 1;\ 1\ 2]$.
- Steady: $Ku = F$; Transient: $M\dot{u} + Ku = F$, θ-scheme stable unconditionally for $\theta \ge \tfrac{1}{2}$.
- Stokes flow: mixed (V, p) system needs LBB-stable elements (Taylor–Hood) or penalty method.

Important Questions

1. What is a functional? Derive the Euler–Lagrange equation for $I = \int (\tfrac{1}{2} k u'^2 - f u)dx$.
2. Explain Rayleigh–Ritz and Galerkin methods with an example; compare MWR techniques.
3. Derive element stiffness matrix for a 1-D linear element; assemble and solve a 2-element problem.

4. Formulate the Galerkin FEM for Stokes flow / potential flow; explain LBB condition.
5. Derive $M\dot{u} + Ku = F$ for transient diffusion and discuss the θ-method and its stability.